

TE Curve Verification Pipeline Familywise ONNX Report - rcim_track1

TE Curve Verification Familywise ONNX Diagnostics

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates `rcim_track1` paper-reference model-bank archives. Each surface loads the selected harmonic amplitude/phase ONNX components from `models/`, reconstructs full TE curves, and compares those curves against dataset-matched held-out measured TE traces.

Unlike standard familywise model-development exports, `rcim_track1` uses a component bank rather than one ONNX file per surface. The surface tables therefore list archive roots and inventory paths; every exact component ONNX and Python path is recorded in the component inventory CSV. The surface path table intentionally uses archive-root glob patterns because one `rcim_track1` surface is assembled from 19 component ONNX models rather than from a single surface-level ONNX file. The harmonic reconstruction applies the paper-faithful `h0` sign convention per curve direction: forward curves use `-1`, backward curves use `+1`.

Output Artifacts

- output directory: `output/validation_checks/track2_familywise_onnx_report/rcim_track1/2026-07-20-13-15-52__track2_rcim_track1_polished_setpoints_familywise_onnx_report` ;
- summary YAML: `output/validation_checks/track2_familywise_onnx_report/rcim_track1/2026-07-20-13-15-52__track2_rcim_track1_polished_setpoints_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV: `output/validation_checks/track2_familywise_onnx_report/rcim_track1/2026-07-20-13-15-52__track2_rcim_track1_polished_setpoints_familywise_onnx_report/model_inventory.csv` ;
- component model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/rcim_track1/2026-07-20-13-15-52__track2_rcim_track1_polished_setpoints_familywise_onnx_report/component_model_inventory.csv` ;
- per-curve metrics CSV: `output/validation_checks/track2_familywise_onnx_report/rcim_track1/2026-07-20-13-15-52__track2_rcim_track1_polished_setpoints_familywise_onnx_report/per_curve_metrics.csv` .

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: rcim_track1 ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	rcim_track1_polished_setpoints_fw	2026-07-13-21-11-39_rcim_track1_polished_setpoints_fw_rcim_track1_polished_input_mode_campaign_validation	polished_setpoint_curve_v1
backward	rcim_track1_polished_setpoints_bw	2026-07-13-21-11-39_rcim_track1_polished_setpoints_bw_rcim_track1_polished_input_mode_campaign_validation	polished_setpoint_curve_v1
global	rcim_track1_polished_setpoints_global	2026-07-13-21-11-39_rcim_track1_polished_setpoints_global_rcim_track1_polished_input_mode_campaign_validation	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/paper_reference/rcim_track1/setpoints/forward/*/onnx/*/*.onnx	models/polished_dataset/paper_reference/rcim_track1/setpoints/forward/*/python/*/*.pk1
backward	models/polished_dataset/paper_reference/rcim_track1/setpoints/backward/*/onnx/*/*.onnx	models/polished_dataset/paper_reference/rcim_track1/setpoints/backward/*/python/*/*.pk1
global	models/polished_dataset/paper_reference/rcim_track1/setpoints/global/*/onnx/*/*.onnx	models/polished_dataset/paper_reference/rcim_track1/setpoints/global/*/python/*/*.pk1

Aggregate Metrics

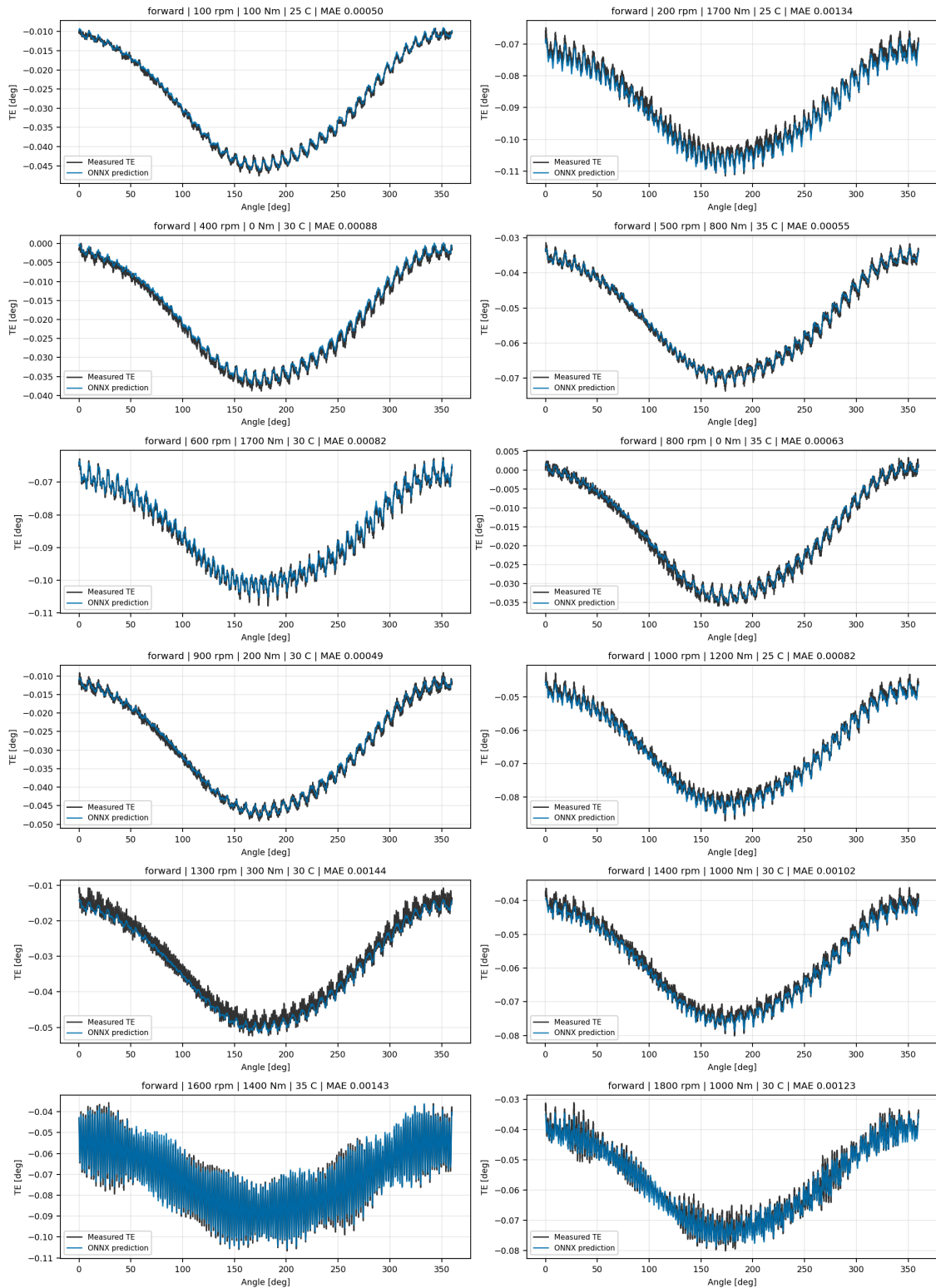
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.001169	0.001389	2.527	5.814
backward	97	0.003582	0.003794	8.356	42.764
global	194	0.002490	0.002756	5.218	21.742

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000111	0.000769	0.000850	0.003634
backward	0.002612	0.003175	0.000970	0.002682
global	0.001207	0.001837	0.001174	0.004152

Forward 12-Curve Page

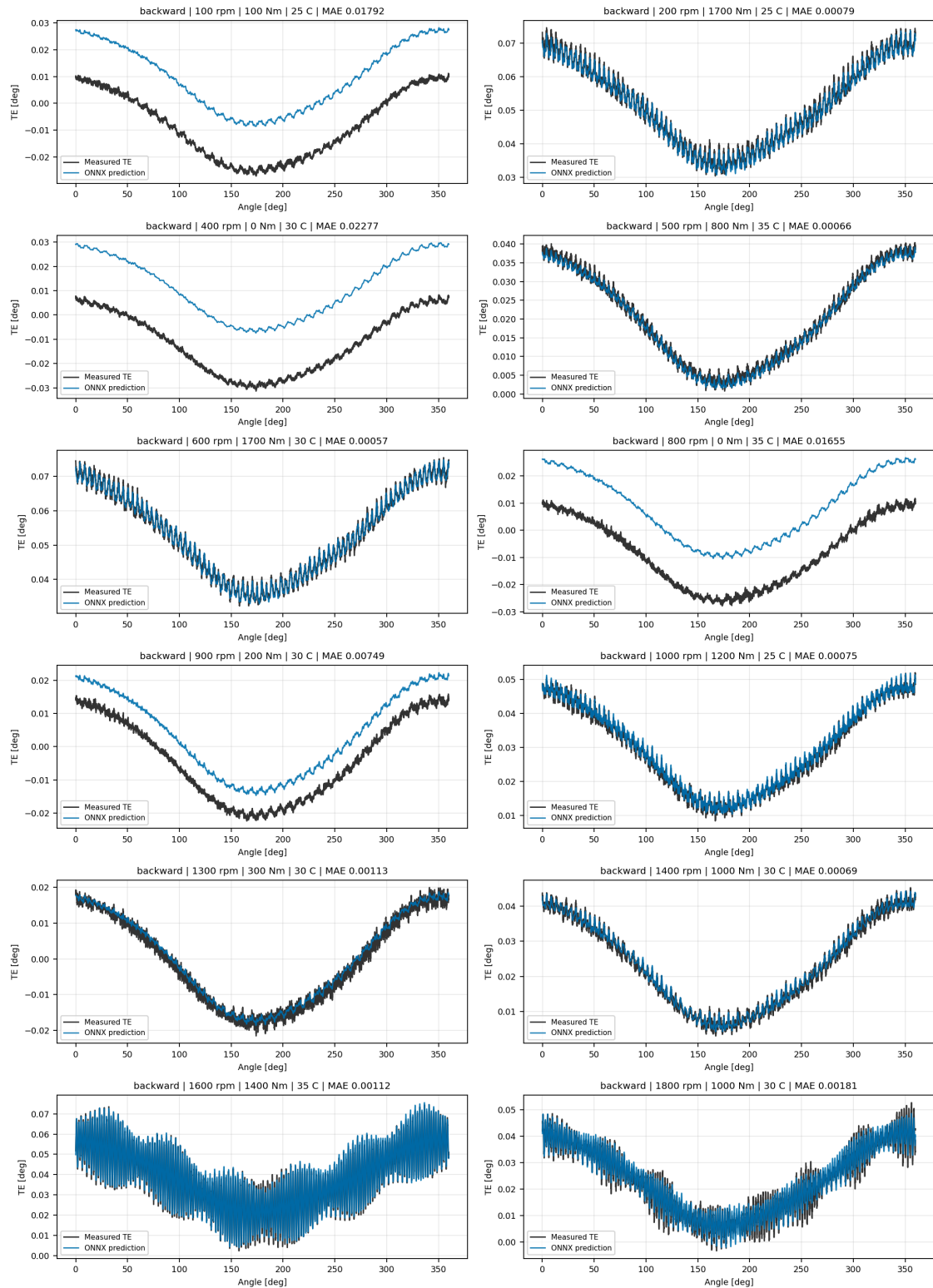
Polished Dataset + Setpoints | rcim_track1 | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

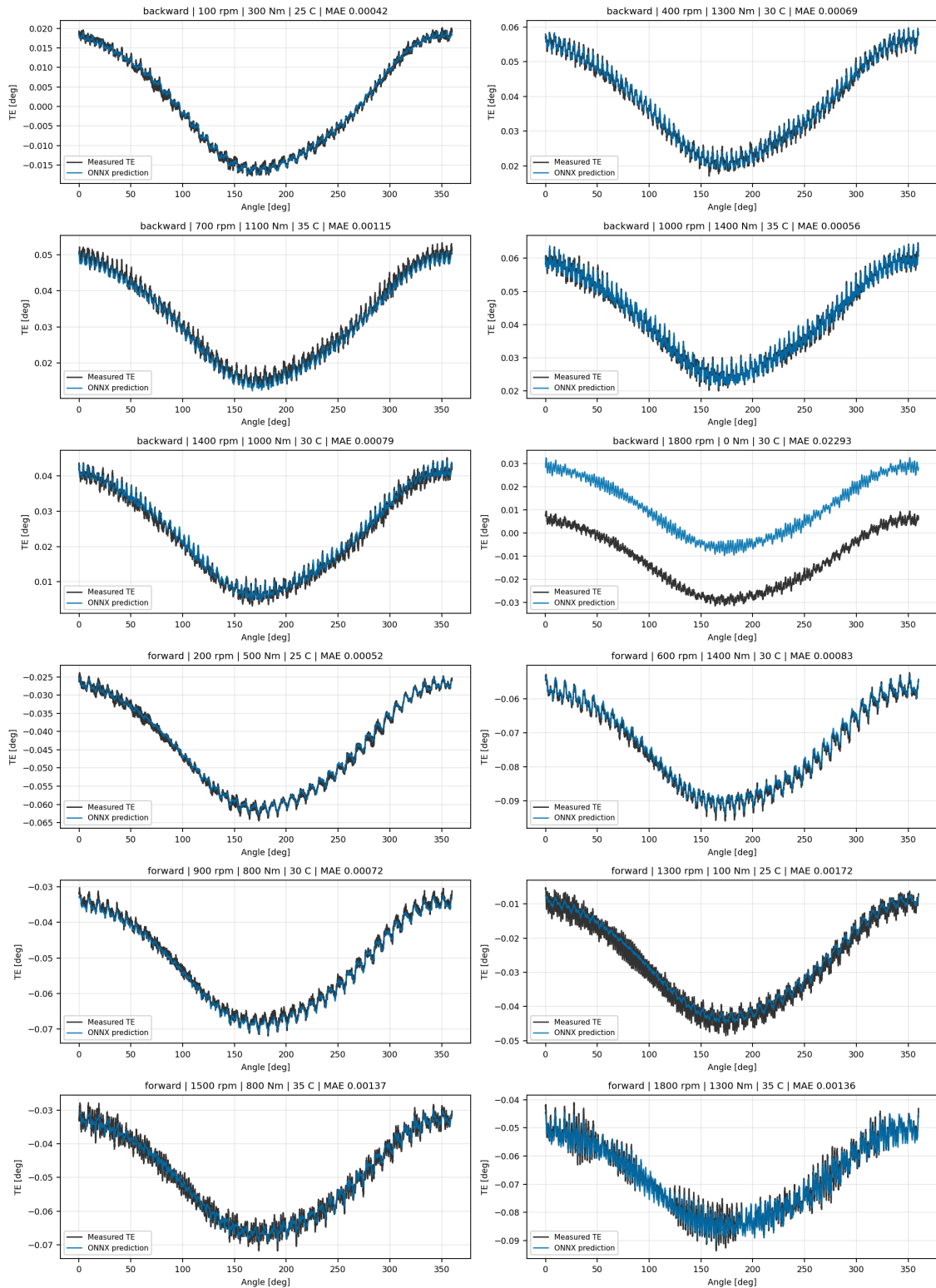
Polished Dataset + Setpoints | rcim_track1 | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | rcim_track1 | global



polished_dataset_setpoints global 12-curve collage